

# JAVA LAB RECORD

Experiment No: 05

Date: 21/09/26

**Title:** Implementation of Deque Interface Methods using ArrayDeque in Java

## 1. Objective / Aim

To implement and demonstrate commonly used methods of the Java **Deque** interface using **ArrayDeque**, including insertion, retrieval, and removal operations from both the front and rear ends.

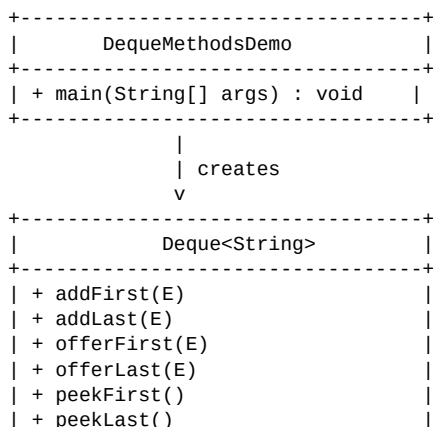
## 2. System Requirements

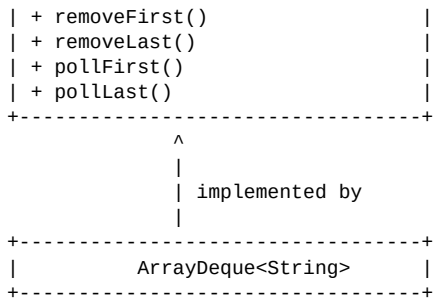
| Hardware Requirements              | Software Requirements                             |
|------------------------------------|---------------------------------------------------|
| Processor: Intel Core i3 or above  | Operating System: Windows / Linux / macOS         |
| RAM: Minimum 4 GB                  | JDK: Java Development Kit 17 or above             |
| Storage: Minimum 500 MB free space | IDE: Eclipse / IntelliJ IDEA / NetBeans / VS Code |
| Keyboard and monitor               | Command Prompt / Terminal                         |

## 3. Algorithm / Logic Description

1. Import the **Deque** interface and **ArrayDeque** class.
2. Create a Deque of type String using **ArrayDeque**.
3. Insert elements at the front using **addFirst()**.
4. Insert elements at the rear using **addLast()**.
5. Insert elements at the front and rear using **offerFirst()** and **offerLast()**.
6. Retrieve the first and last elements without removing them using **peekFirst()** and **peekLast()**.
7. Remove elements from the front and rear using **removeFirst()** and **removeLast()**.
8. Remove elements safely from the front and rear using **pollFirst()** and **pollLast()**.
9. Display the Deque after each operation.
10. Compile and execute the program and verify the output.

## 4. UML Class Diagram





## 5. Source Code

```
import java.util.Deque;
import java.util.ArrayDeque;

class DequeMethodsDemo {

    public static void main(String[] args) {

        // Creating a Deque
        Deque<String> items = new ArrayDeque<>();

        // addFirst()
        items.addFirst("Book");
        items.addFirst("Pen");

        System.out.println("After addFirst(): " + items);

        // addLast()
        items.addLast("Bag");
        items.addLast("Bottle");

        System.out.println("After addLast(): " + items);

        // offerFirst()
        items.offerFirst("Notebook");

        System.out.println("After offerFirst(): " + items);

        // offerLast()
        items.offerLast("Pencil");

        System.out.println("After offerLast(): " + items);

        // peekFirst()
        System.out.println("First element: " + items.peekFirst());

        // peekLast()
        System.out.println("Last element: " + items.peekLast());

        // removeFirst()
        System.out.println("Removed first: " + items.removeFirst());

        System.out.println("After removeFirst(): " + items);

        // removeLast()
        System.out.println("Removed last: " + items.removeLast());

        System.out.println("After removeLast(): " + items);

        // pollFirst()
        System.out.println("Polled first: " + items.pollFirst());

        System.out.println("After pollFirst(): " + items);

        // pollLast()
        System.out.println("Polled last: " + items.pollLast());

        System.out.println("After pollLast(): " + items);

    }
}
```

## 6. Completion and Execution Commands

Step 1: Save the source file as:

DequeMethodsDemo.java

Step 2: Open Command Prompt / Terminal and move to the folder containing the file.

Step 3: Compile the program:  
javac DequeMethodsDemo.java

Step 4: Execute the program:  
java DequeMethodsDemo

The program should compile successfully and display the Deque operations in the terminal.

## 7. Output

```
After addFirst(): [Pen, Book]
After addLast(): [Pen, Book, Bag, Bottle]
After offerFirst(): [Notebook, Pen, Book, Bag, Bottle]
After offerLast(): [Notebook, Pen, Book, Bag, Bottle, Pencil]
First element: Notebook
Last element: Pencil
Removed first: Notebook
After removeFirst(): [Pen, Book, Bag, Bottle, Pencil]
Removed last: Pencil
After removeLast(): [Pen, Book, Bag, Bottle]
Polled first: Pen
After pollFirst(): [Book, Bag, Bottle]
Polled last: Bottle
After pollLast(): [Book, Bag]
```

## 8. Result / Completion

The Java program was successfully compiled and executed. The Deque operations **addFirst()**, **addLast()**, **offerFirst()**, **offerLast()**, **peekFirst()**, **peekLast()**, **removeFirst()**, **removeLast()**, **pollFirst()**, and **pollLast()** were demonstrated successfully using `ArrayDeque`.